

1 Conclusions

This research successfully developed an end-to-end, multi-tenant Software-as-a-Service (SaaS) platform, transforming isolated deep learning algorithms into an industrial power grid defect inspection system.

1.1 Summary of Achievements

- **High-Precision Detection via Deep Learning:** Successfully integrated Mask R-CNN and YOLOv8s. Despite constrained datasets, targeted optimisation enabled accurate segmentation of complex bird nests and high-precision bounding over oil leak signatures.
- **Hybrid Thermal Diagnostics:** Engineered a cost-effective hybrid infrared pipeline. By combining EasyOCR with dynamic pixel-to-temperature mappings, the system flags critical overheating anomalies in approximately 765 ms without relying on dedicated, resource-heavy thermal neural models.
- **Robust SaaS Engineering:** Delivered a Flask-driven RESTful architecture overcoming fundamental hardware limitations. The event-driven `ThreadPoolExecutor` and lazy-loaded Model Registry sustained processing of over 120 images per minute under 12GB VRAM constraints. Explicit SQL predicate injection guaranteed zero cross-tenant data leakage.

1.2 Limitations

- **Environmental Variance:** Heavily augmented training data remains biased toward ideal lighting, reducing accuracy in extreme weather or night operations.
- **OCR Sensitivity:** The infrared mapping module is vulnerable to legacy camera watermarks and low-resolution compression, which can occasionally defeat heuristic masking and degrade temperature readings.
- **Scalability Limits:** Local thread-pool deployment restricts horizontal scaling required for thousands of concurrent enterprise requests.

1.3 Future Work

Future iterations will prioritise transitioning visual detection backbones to multi-modal architectures like Vision Transformers (ViT) to increase contextual robustness against weather anomalies. Furthermore, migrating the SaaS backend to a cloud-native Kubernetes (K8s) environment will unlock distributed GPU auto-scaling, completing the transition to nationwide industrial adoption.